

UPIPaymentAndDigitalWalletSystem

Java Programming Laboratory Experiment

Experiment No.	04
Date	16/09/2026
Title	UPI Payment and Digital Wallet System

Objective / Aim

To design and implement a UPI Payment and Digital Wallet System in Java using interfaces, classes, objects, encapsulation, custom exceptions, exception handling, method overriding, and the finally block. The system supports adding money, checking balance, making UPI payments, and handling invalid UPI IDs, invalid amounts, and insufficient balance.

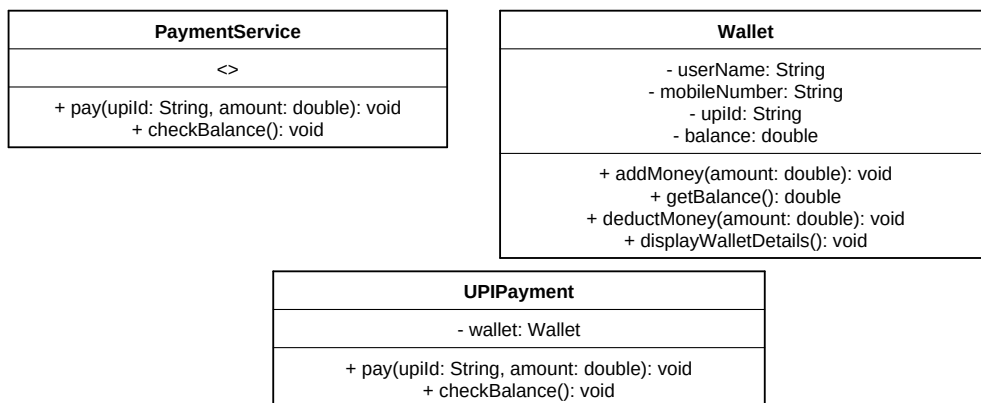
System Requirements

HARDWARE	SOFTWARE
- Standard PC/Laptop - Intel Core i3/i5 processor or equivalent - Minimum 4 GB RAM - Minimum 500 MB free storage	- Operating System: Windows / Linux / macOS - Java Development Kit (JDK 17 or JDK 21) - Text Editor / IDE: Eclipse, IntelliJ IDEA, NetBeans, or VS Code - Command Prompt / Terminal

Algorithm / Logical Description

1. Create a PaymentService interface with pay() and checkBalance() methods.
2. Create custom exceptions for insufficient balance, invalid UPI ID, and invalid amount.
3. Create a Wallet class with private user, mobile, UPI ID, and balance data members.
4. Define methods to add money, get balance, deduct money, and display wallet details.
5. Create UPIPayment class implementing PaymentService.
6. Validate UPI ID and payment amount before processing a transaction.
7. Check whether sufficient wallet balance is available.
8. Throw and catch the appropriate custom exception when validation fails.
9. Use finally to display transaction completion.
10. Create the wallet and perform valid, invalid, and insufficient-balance payment tests.

UML Class Diagram



Relationship: UPIPayment implements PaymentService and uses a Wallet object. Wallet encapsulates wallet information and balance operations. Custom exception classes handle transaction validation errors.

Source Code

```
package javacore;
public class MainUPIPayment { // PaymentService Interface
    static interface PaymentService {

        void pay(String upId, double amount);

        void checkBalance();
    }

    // Custom Exception for insufficient balance
    static class InsufficientBalanceException extends Exception {

        public InsufficientBalanceException(String message) {
            super(message);
        }
    }

    // Custom Exception for invalid UPI ID
    static class InvalidUPIException extends Exception {

        public InvalidUPIException(String message) {
            super(message);
        }
    }

    // Custom Exception for invalid amount
```

```

static class InvalidAmountException extends Exception {
    public InvalidAmountException(String message) {
        super(message);
    }
}

// Wallet Class
static class Wallet {
    // Private data members to demonstrate encapsulation
    private String userName;
    private String mobileNumber;
    private String upiId;
    private double balance;

    // Constructor
    public Wallet(String userName, String mobileNumber, String upiId) {
        this.userName = userName;
        this.mobileNumber = mobileNumber;
        this.upiId = upiId;
        this.balance = 0.0;
    }

    // Method to add money
    public void addMoney(double amount) {
        if (amount > 0) {
            balance = balance + amount;
            System.out.println("Money added successfully: ." + amount);
        } else {
            System.out.println("Invalid amount.");
        }
    }

    // Method to get balance
    public double getBalance() {
        return balance;
    }

    // Method to deduct money
    public void deductMoney(double amount) {
        balance = balance - amount;
    }

    // Method to display wallet details
    public void displayWalletDetails() {
        System.out.println("\n----- Wallet Details -----");
        System.out.println("User Name : " + userName);
        System.out.println("Mobile Number: " + mobileNumber);
        System.out.println("UPI ID : " + upiId);
        System.out.println("Balance : ." + balance);
    }
}

// UPI Payment Class
static class UPIPayment implements PaymentService {
    private Wallet wallet;

    // Constructor
    public UPIPayment(Wallet wallet) {
        this.wallet = wallet;
    }

    // Pay method
    @Override
    public void pay(String upiId, double amount) {
        try {
            // Validate UPI ID
            if (upiId == null ||
                !upiId.contains("@") ||
                upiId.startsWith("@") ||
                upiId.endsWith("@")) {
                throw new InvalidUPIException(
                    "Invalid UPI ID: " + upiId
                );
            }

            // Validate payment amount
            if (amount <= 0) {
                throw new InvalidAmountException(
                    "Payment amount must be greater than zero."
                );
            }

            // Check sufficient balance
            if (amount > wallet.getBalance()) {

```

```

        throw new InsufficientBalanceException(
            "Insufficient balance for this payment."
        );
    }

    // Deduct payment amount
    wallet.deductMoney(amount);

    System.out.println("\nPayment Successful!");
    System.out.println("Receiver UPI ID: " + upiId);
    System.out.println("Amount Paid      : ." + amount);
}

catch (InvalidUPIException e) {
    System.out.println("Transaction Failed: "
        + e.getMessage());
}

catch (InvalidAmountException e) {
    System.out.println("Transaction Failed: "
        + e.getMessage());
}

catch (InsufficientBalanceException e) {
    System.out.println("Transaction Failed: "
        + e.getMessage());
}

finally {
    System.out.println("Transaction processing completed.");
}
}

// Check balance method
@Override
public void checkBalance() {
    System.out.println(
        "Available Wallet Balance: ."
        + wallet.getBalance()
    );
}
}

```

Completion and Execution Commands

Save the source code exactly as:	Main.java
Compile the program:	javac Main.java
Run the program:	java Main

Output Screen

```
----- Wallet Details -----
User Name      : Sameera
Mobile Number: 9876543210
UPI ID         : sameera@upi
Balance        : ₹0.0
Money added successfully: ₹5000.0
Available Wallet Balance: ₹5000.0

--- Valid Payment ---

Payment Successful!
Receiver UPI ID: rahul@upi
Amount Paid      : ₹1500.0
Transaction processing completed.
Available Wallet Balance: ₹3500.0

--- Invalid UPI Test ---
Transaction Failed: Invalid UPI ID: rahulupi
Transaction processing completed.

--- Invalid Amount Test ---
Transaction Failed: Payment amount must be greater than zero.
Transaction processing completed.

--- Insufficient Balance Test ---
Transaction Failed: Insufficient balance for this payment.
Transaction processing completed.

----- Wallet Details -----
User Name      : Sameera
Mobile Number: 9876543210
UPI ID         : sameera@upi
Balance        : ₹3500.0
```

Results

The UPI Payment and Digital Wallet System was successfully implemented in Java. The program demonstrates interface implementation, encapsulation, custom exception handling, validation of UPI payments, balance checking, successful transactions, and use of the finally block for transaction completion.